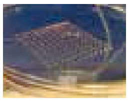
New 3D Mark released
Futuremark's popular benchmarking software - 3D Mark now has a new version. Get it for your rig here **http://dgit.in/WrBlSC**



3D printed Stem cells
A team at a University in Edinburgh, Scotland have developed a method to 3D print human embryonic cells **http://dgit.in/11p0qDJ**

HOW WE TESTED

Solid state drive testing is slightly different from your regular HDD testing. To ensure that there were absolutely no bottlenecks in the testing and to give you a fair idea of the real world transfer speeds, we used the following test setup:

- **Processor:** Intel Core i7-3700K @ 3.5GHz
- **RAM:** 2 x 4GB Kingston HyperX RAM
- **Motherboard:** ASRock Z77 Extreme 4
- **OS Drive:** Kingston HyperX 120 GB
- **RAID 0 drives:** 2x Corsair Neutron GTX 240 GB drives
- **Graphics Card:** AMD HD 6850
- **Power Supply Unit:** Corsair HX 620W

We had a healthy mix of synthetic tests and real world tests. Before testing, we securely erased all the SSDs using a utility called DiskWipe. This free utility securely erases any data that may be present on the SSD. We used the One Pass Zeros erasing pattern which writes zeroes on the entire SSD. This ensures that the SSD is completely free of any content. This helped prevent putting the SSDs that may have been used in the past at a disadvantage. Also all drives were connected to the SATA 6 Gbps ports on the ASRock Z77 Extreme 4 motherboard.

Crystal Disk Mark is a renowned utility known for testing drives and helps measure sequential and random read/write speeds. We kept the size at 1000MB and ran five iterations of the test, so as to get a fair score.

AS-SSD is another benchmark we used to test the sequential and random read/write scores. In the random tests the file size is around 4KB. This test also gives us the access time which is the time required to access data from the various memory chips.

The final synthetic test that we ran was PC Mark 7's secondary storage test. This test stresses the SSD by simulating workloads such as adding music, importing pictures, video editing etc., to give an overall score at the end. Higher score implies better SSD.

Synthetic tests just give you a part of the whole picture. While the scores certainly give you an idea of the transfer speeds, they are in no way indicative of the fact that you will get the same transfer speeds in real life scenarios. We had two test files - an 8GB sequential file and an 8 GB assorted file (which has a random mix of files ranging in capacity from a few KBs to a few GBs). These files were transferred from the RAID 0 array to the SSD file under test to simulate a write operation; from the SSD to RAID 0 array to simulate a read and finally between the two partitions of the SSD to simulate intra-drive transfer rate. The system was restarted between each of these three transfer tests, so as to get rid of any buffer.

As solid state drives are all about performance, we did not have any weightage for features, as there are none so to speak. Ultimately everything boils down to transfer speeds and pricing.

channel controller, there are only four NAND chips (4x64GB 19nm SanDisk eX2 ABL). Apart from this, there is a Samsung DDR2-800 caching DRAM onto which data is temporarily cached before finally moving to and from the NAND chips. The NAND chips have a special feature called nCache which makes part of the eX2 ABL multi-level-cell (MLC) NAND act as non-volatile single-level cell (SLC) NAND called as nCache. Generally in an SSD, as the incompressible file size goes below 1MB, the performance does tend to take a hit. In order to prevent smaller sized incompressible data from populating the MLC NAND, it is housed in the nCache. The nCache being non-volatile in nature, has a lifespan which is 10 times that of MLC NAND.

Thus nCache tends to increase the overall endurance of the SSD while optimizing performance during incompressible data writes.

Corsair has been quite consistent as far as SSDs in the Indian market go. At Computex 2012, Corsair announced an exclusive partnership with Link-A-Media Devices, a new controller maker in the consumer SSD space. This gave rise to the Neutron and Neutron GTX series of drives last year. While the Neutron GTX 240GB drive won the Best Performer award last year, it was excluded from this test, but we still have it in the comparison table to give you an indication of how good or bad the current crop of SSDs is. The Neutron 240GB comes in a 7mm profile grey-coloured enclosure, opening which you will find the LM87800 controller. Eight synchronous MLC NAND modules are present on each side of the PCB along with a 128 GB Samsung DDR2 cache buffer on each side.

The other Corsair SSD present in the test was the red-coloured SandForce SF2281 controller sporting Corsair Force